# Preliminaries and Review

## Velocity and Acceleration

**Question** Suppose that a particle is moving along a straight line, with position given by a function $s(t)$. Then _____ gives its velocity and _____ its acceleration. Suppose $s(t) = t^3 - 5t^2$ gives its position in meters to the right of a reference point at time $t$ seconds. Find the particles position, velocity, and acceleration at $t = 2$ seconds. Include units. Where is it relative to the reference point at that time? In which direction is it moving? Is it speeding up or slowing down?

## Differential Equations

**Question** By antidifferentiating, solve the initial value problem

$$s''(t) = -6t,\ s'(1) = 0,\ s(2) = 3.$$

# From Acceleration to Position

As we noted above, knowing the acceleration of an object at time $t$ tells about the _____ derivative of its position. By combining this idea with our solutions of differential equations, we can compute a particle's position function given a formula for its acceleration at time $t$ and some information about its position and/or its velocity at some given time(s).

For the first two questions, consider the following description:

> Suppose that an object moving along a straight line has constant acceleration $a(t) = 2$ meters per second per second. Suppose also that the object is 2 meters to the right of a reference point at time 0 seconds, moving to the left at 3 meter per second. Take motion to the right as positive.

1. □Regrade (a) By setting up and solving an appropriate initial value problem, find expressions for $s(t)$ and $v(t)$, the position and velocity of the object at time $t$ respectively.

(b) Fill the blanks: At time 0 seconds, the object is ___ meters to the _____ of the reference point. It is moving to the _____ at a speed of ___ meters per second.

(c) When does the object come to momentary stop? Where is it at that point in time?

(d) When does the particle pass the reference point? How fast is it going? In which direction?

2. □Regrade (a) Fill in the blanks: initially, the particle travels _____ for _____ seconds. It passes the reference point after ____ second, moving at ____ m/sec. At time _____ seconds, it comes to a momentary halt _____ meters to the _____ of the reference point. After that, it moves to the _____, passing the reference point again at time ____ seconds, moving at ____ m/sec.

(b) Before it turns around, is the particle speeding up or slowing down? What about after that time? Explain your different answers, given that acceleration is always a constant 2 m/sec$^2$. Consider what the initial velocity is, and what happens to the velocity each second.

3. □Regrade Suppose that a particle moving along a straight line has acceleration $a(t) = 5$ ft/sec$^2$ and that the particle starts two feet to the right of a reference point with initial velocity $v(0) = -10$ ft/sec. Describe the subsequent motion of the particle. (Hint: First use the differential equation and the initial conditions to find $s(t)$. Then describe the motion similarly to Questions 1 and 2. Any decimals should be written to two decimal places.)

4. A particle moves on a line with acceleration $a(t) = -4t$. Suppose that $s(0) = 6$ and $v(0) = 18$. Find the functions $s(t)$ and $v(t)$. Describe the motion in words for time $t \geq 0$ seconds. Assume distance is measured in meters.

□Regrade

(Hint: $s(t)$ turns out to be a cubic. Feel free to use Geogebra/Desmos to find its roots.)

5. Recall that Newton's First Law of Motion says $F = ma$, where

□Regrade

- $F$ is _____, measured in _____.
- $m$ is _____, measured in _____.
- $a$ is _____, measured in _____.

In textbooks it is often stated that "Newton's first law says that objects will move with constant velocity if the force applied to them is zero." Explain why this follows from Newton's Law of Motion.

(Hint: if the force is zero, what can you say about acceleration? Set up and solve a differential equation for velocity.)